

Closed-Book Knowledge Base Construction with Calibrated Abstention and Numeric Self-Consistency

Tanvir Ahmed

North South University, Dhaka, Bangladesh

`tanvir.ahmed32@northsouth.edu`

Abstract

Large language models (LMs) store a remarkable amount of factual knowledge in their parameters, yet turning that knowledge into a reliable, *set-valued* knowledge base—where a subject may have zero, one, or many objects—remains difficult. The LM-KBC 2026 shared task (5th edition, at the AKBC workshop, EMNLP 2026) poses exactly this in a strict *closed-book* setting: an open-weight model of at most 32B parameters, with no retrieval, no external knowledge, and no fine-tuning. We start from the evaluation metric and observe that it rewards *abstention*: predicting the empty set for every instance already scores 0.175 relation-averaged macro- F_1 , close to the official baseline (0.273). We therefore treat *when to answer* as a first-class decision and build a per-relation pipeline that (i) grounds each prompt in the relation’s exact scope, (ii) induces calibrated abstention by *self-consistency voting*—abstaining when repeated samples disagree—rather than by instruction, and (iii) aggregates numeric relations by median. With **Gemma-4-31B** (a 30.7B open-weight model, the strongest in the budget) the system reaches 0.459 relation-averaged macro- F_1 (0.490 micro), beating the official baseline (0.273/0.284) on all six relations; a smaller Qwen2.5-14B run locally in 4-bit reaches 0.416. All generation is cached, so the entire decode stage is reproducible without a GPU. We release our code.¹

1 Introduction

Pre-trained language models (LMs) store a remarkable amount of factual knowledge in their parameters, acquired during pre-training on large text corpora (Petroni et al., 2019; Roberts et al., 2020), and a body of work probes and extracts it (Jiang et al., 2020; Zhong et al., 2021; AlKhamissi et al., 2022). This raises the prospect of building a

structured knowledge base (KB) directly from an LM rather than from curated sources (Dong et al., 2014; Vrandečić and Krötzsch, 2014). Doing so reliably is harder than answering a single factual question: a KB entry is a *complete set* of facts, so the model must decide not only what is true but how many objects—possibly none—a subject and relation admit.

The LM-KBC 2026 task. The shared task (Ali-vanistos et al., 2022; Kalo et al., 2023, 2025) formalises this as: given a subject s and relation r , predict the complete object set $\{o_1, \dots, o_k\}$, where k may be zero, one, or many. Predictions are scored by macro precision, recall, and F_1 , with string normalisation and alias matching for entities and a 5% tolerance for numeric relations. The 2026 edition is strictly *closed-book*: a system may use any open-weight model up to 32B parameters, but no web search, no retrieval, no external corpus, and no fine-tuning. This rules out the retrieval and tuning that strong entries in earlier editions relied on (Zhang et al., 2023; Asai et al., 2024), leaving parametric recall and *prompt design* as the only levers.

Why this is hard. Unlike single-answer probing benchmarks such as LAMA (Petroni et al., 2019), which assume exactly one gold object and reward ranking it highly, this setting demands the *entire* set—and the empty set is a legitimate, frequent answer (a living person has no recorded city of death; an island country has no land border). Three difficulties follow. First, LM outputs are stochastic and prompt-sensitive, so a single query yields unstable sets. Second, relations span very different cardinalities, from a single number to hundreds of award recipients, complicating aggregation. Third, the model must recognise when it does not know an answer and *abstain* rather than fabricate one.

¹<https://github.com/ANON/lm-kbc-2026>

Our approach. We start from the evaluation metric itself. Because macro- F_1 scores an empty prediction against an empty gold set as perfect, simply predicting nothing everywhere already attains 0.175 relation-averaged macro- F_1 on validation—competitive with the official baseline (0.273) and better on the most null-heavy relations (Section 3). The lesson is that points come not from answering everything, but from recalling the few numeric values and, above all, knowing when *not* to answer. We therefore make abstention a first-class decision: we sample each null-heavy relation several times and keep an object only when the samples agree, collapsing to the empty set when the model is internally inconsistent. Scope-grounded prompts and median aggregation for numeric relations complete the pipeline (Figure 1).

Our contributions are:

1. A metric analysis (Section 3) showing that, under the macro metric, abstention is a high-value action: the predict-nothing baseline already rivals the official baseline, which reframes the task around *when to answer*.
2. A per-relation pipeline (Figure 1, Section 4) that grounds prompts in exact scopes and induces *calibrated abstention* via self-consistency voting, with no trained confidence model: it lifts a weak 0.5B model from 0.173 to 0.372 on the null-heavy relations and still helps a strong 14B.
3. A reproducible open-weight system reaching 0.459 relation-averaged macro- F_1 with Gemma-4-31B (0.416 with a local 4-bit Qwen2.5-14B), beating the official baseline on all six relations, with cached generation enabling GPU-free decode-stage tuning (Sections 5–7).

2 Background and Related Work

LMs as knowledge bases. Petroni et al. (2019) cast factual recall as cloze probing; follow-ups sharpened the prompts (Jiang et al., 2020; Shin et al., 2020; Zhong et al., 2021), quantified how much knowledge a model’s parameters hold (Roberts et al., 2020), and questioned how reliably it can be read out (Cao et al., 2021; AlKhamissi et al., 2022). These probes assume a single gold object and reward ranking it highly. We instead target the set-valued, alias-graded variant in which the empty set is a valid answer, which is closer to

materialising an actual KB (Pan et al., 2023).

The LM-KBC shared task. Across its editions (Alivanistos et al., 2022; Kalo et al., 2023, 2024, 2025) the task has moved from masked-LM probing and prompt ensembling toward instruction-tuned generative systems, with some earlier tracks permitting retrieval or fine-tuning (Zhang et al., 2023). The 2026 edition’s closed-book, no-tuning, ≤ 32 B rules remove those options—which is precisely what makes a parametric-recall-plus-abstention design necessary.

Retrieval vs. closed book. The usual route to knowledge-intensive accuracy is retrieval augmentation (Lewis et al., 2020; Guu et al., 2020), which helps most exactly when parametric memory is weak (Mallen et al., 2023). Self-RAG (Asai et al., 2024) even lets the model decide *when* to retrieve; we borrow the spirit of a model-driven, adaptive decision but, with retrieval forbidden, apply it to a different one—*when to answer at all*, not what to fetch.

Self-consistency, calibration, and abstention. These threads converge in our method. Self-consistency (Wang et al., 2023; Chen et al., 2023) samples a model repeatedly and aggregates the generations to pick a single answer. Abstaining when uncertain is the classic reject option (Geifman and El-Yaniv, 2017), studied in NLP as calibration (Guo et al., 2017), unanswerable QA (Rajpurkar et al., 2018), and selective QA (Cole et al., 2023), and LMs retain some awareness of what they know (Kadavath et al., 2022). We combine the two: cross-sample *disagreement* (Kuhn et al., 2023; Manakul et al., 2023) is our zero-parameter confidence signal, and—unlike majority-vote self-consistency, which always emits an answer—we let it trigger *abstention*, because here the empty set is itself a high-value prediction.

3 Task and Metric Analysis

Relations. The dataset comprises six Wikidata relations (Table 1) that fall into three groups. *Numeric* relations (HASCAPACITY, HASAREA) have a single scalar object; *large-set* AWARD-WONBY has up to hundreds of recipients per subject; and three *null-heavy* string relations (COUNTRYLANDBORDERSCOUNTRY, PERSONHASCITYOFDEATH, COMPANYTRADESAT-STOCKEXCHANGE) leave a large fraction of subjects with no valid object at all. Each rela-

Relation	#val	empty%
COUNTRYLANDBORDERSCOUNTRY	68	25
PERSONHASCITYOFDEATH	100	45
COMPANYTRADESATSTOCKEXCHANGE	100	35
AWARDWONBY	10	0
HASCAPACITY	100	0
HASAREA	100	0

Table 1: Validation statistics. AWARDWONBY averages ~ 148 objects per subject (max 641); three string relations are often legitimately empty.

tion carries a precise scope—land-only borders, city-granular death location, the highest published capacity—which we copy verbatim into the prompt so the model targets the same notion the gold set was built against.

Evaluation. Predictions are scored against the gold object set by macro precision and recall. A predicted string is a true positive if its normalised form (lower-cased, diacritics and punctuation removed) matches any alias of a gold object, and a numeric prediction if it lies within 5% of the gold value. Two boundary conventions drive everything that follows: precision is defined as 1 when the prediction is empty, and recall as 1 when the gold set is empty. We report *relation-averaged* macro- F_1 (the mean of the six per-relation scores) and *micro-averaged* F_1 (over all instances).

The metric makes abstention valuable. Those two conventions make an empty prediction against an empty gold set a perfect F_1 , with a large consequence: predicting the empty set for *every* validation instance already scores the “Empty” bars in Figure 3, averaging 0.175 relation-averaged macro- F_1 (0.203 micro). This is on par with the official baseline’s 0.273 and *above* it on the two most null-heavy relations (PERSONHASCITYOFDEATH 0.45 vs. 0.16; COMPANYTRADESATSTOCKEXCHANGE 0.35 vs. 0.32). The task therefore rewards not answering everything, but two narrower abilities: recalling the few numeric values, and abstaining precisely when no object exists. These are the levers our pipeline pulls.

4 System

Architecture overview. The pipeline (Figure 1) maps each (s, r) to one chat prompt, decodes N or n cached samples, and aggregates them per relation kind into an object set. It has no trainable

Algorithm 1 CALIBRATEDABSTENTION

Require: samples $S = \{s_1, \dots, s_N\}$, threshold θ , *multi*
1: $c[o] \leftarrow \#\{s_i : o \in \text{PARSE}(s_i)\}$ for each object o
2: **if** *multi* **then return** $\{o : c[o]/N \geq \theta\}$
3: **else**
4: $o^* \leftarrow \arg \max_o c[o]$; **return** $c[o^*]/N \geq \theta ? \{o^*\} : \emptyset$
5: **end if**

parameters and targets the two analysis levers: abstention and numeric recall.

Scope-grounded prompts. Each relation’s system message states the task, the *exact* ground-truth scope (copied verbatim from the official definitions), and the required answer form (Table 2); the model emits one JSON object $\{\text{"answers"} : [\dots]\}$ (Figure 2), which makes the empty case unambiguous. Few-shot exemplars come from the training split; for null-heavy relations we interleave empty and non-empty exemplars so abstention is demonstrated, not merely instructed. Even a 0.5B model emits valid JSON on 100% of instances, so format is never a bottleneck.

Calibrated abstention via self-consistency. Scope grounding tells the model *what* counts as an answer, but not *when* to stay silent. A bare instruction to “answer empty when unsure” is unreliable: an unconstrained small model names a death-city for living people and invents stock exchanges, dropping precision *below* the empty baseline (Section 7). Instead, for the three null-heavy relations we sample $N=3$ completions at temperature $\tau=0.7$ and keep an object only if it appears in at least a fraction θ of samples; single-object relations abstain entirely if even the top answer is below θ (Algorithm 1). When the model guesses, samples disagree and the prediction collapses to $\emptyset$ —where the metric rewards silence. This is disabled for AWARDWONBY (recall-bound) and the numeric relations.

Numeric aggregation. The numeric relations need a different aggregator. HASCAPACITY and HASAREA take a single number graded within 5%. We sample $n=5$ completions and take the *median* of the parsed numbers (Wang et al., 2023) for robustness.

Robust parsing. Whatever the relation kind, every sample passes through a common final stage. A relation-aware parser recovers the answer list from noisy output (code fences, prose), strips ex-

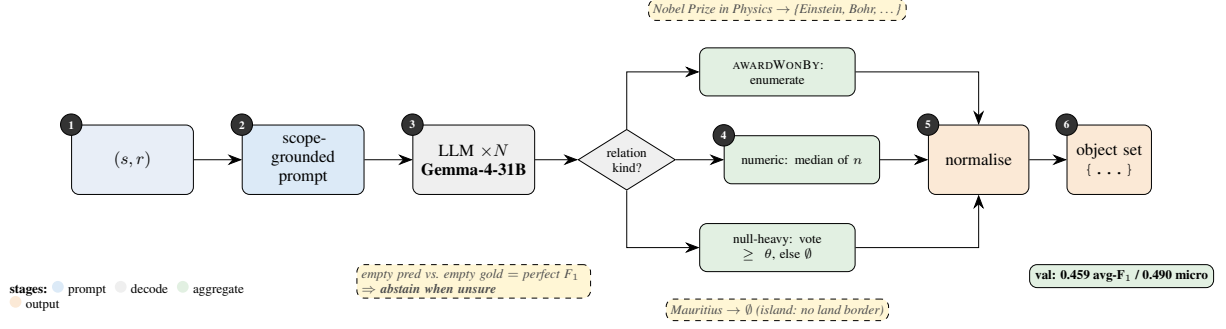


Figure 1: System pipeline (no trainable parameters). Each (s, r) is rendered into a scope-grounded chat prompt (stage 2), the model is sampled N times (stage 3, cached), and the samples are aggregated by relation kind (stage 4): *median* for numeric relations, consistency *voting with abstention* (keep an object only if it appears in $\geq \theta$ of samples, else $\emptyset$) for the null-heavy string relations, and *enumeration* for AWARDWONBY; a final normalisation matches the evaluator exactly. Dashed boxes are worked examples—the island MAURITIUS correctly collapses to $\emptyset$. The design follows directly from the metric insight (lower-left badge). Reported score is Gemma-4-31B on validation.

Relation	Scope stated in the prompt	Answer form
COUNTRYLANDBORDERSCOUNTRY	countries sharing a <i>land</i> border; maritime borders excluded; an island has none	countries, or []
PERSONHASCITYOFDEATH	the <i>city</i> where the subject died; empty if alive or the city is unknown	one city, or []
COMPANYTRADESATSTOCKEXCHANGE	stock exchange(s) on which the company is publicly listed; empty if unlisted	exchanges, or []
AWARDWONBY	every recipient of the award (often tens to hundreds)	names (long)
HASCAPACITY	maximum spectator capacity in people; highest published value	one integer
HASAREA	total surface area in km ² (land + inland water)	one number

Table 2: Per-relation prompt specification. Each system message states the relation’s exact scope (copied from the official definitions) and the required answer form; the model replies with a single JSON object `{"answers": [...]}`.

```

SYS: precise KB engine (closed book).
TASK: city where the subject died.
SCOPE: city granularity; [] if
alive/unknown.
Reply only {"answers":[...]}.
U: Egbert Mulder
A: {"answers":["Groningen"]}
U: Dave Keon
A: {"answers":[]}
U: <query> A:

```

Figure 2: Abbreviated prompt for a null-heavy relation. Mixed exemplars teach abstention; output is forced JSON.

licit empty markers, extracts one number for numeric relations (handling commas, units, scale words), and de-duplicates. Normalisation (lower-casing, diacritic/punctuation removal) is byte-identical to the official evaluator; pushing gold answers back through the parser scores 1.000, confirming it is lossless.

5 Experimental Setup

Models. Our primary system uses **Gemma-4-31B-it** (Gemma Team, 2024), a 30.7B-parameter open-weight model within the 32B cap. Because it exceeds the memory of the single 16 GB GPU available to us, we serve it through a hosted OpenAI-compatible API; the weights are open and no fine-tuning is applied, so the closed-book rules are respected (the API is used purely as compute). For the self-consistency ablations (Section 7) we additionally run **Qwen2.5-14B-Instruct** (Qwen Team, 2024) (14.7B) locally in 4-bit. Several nominally “32B” models in fact exceed the cap (Qwen2.5-32B at 32.5B, Qwen3-32B at 32.8B) and are ineligible.

Decoding. The pipeline selects an aggregator per relation kind (Section 4): median over $n=5$ samples for the numeric relations, consistency voting over $N=3$ samples at threshold $\theta=0.5$ for the null-heavy relations (Algorithm 1), and

a single greedy enumeration for AWARDWONBY. These self-consistency configurations (temperature $\tau=0.7$) are evaluated on Qwen2.5-14B, where we can sample freely. For Gemma-4-31B, API rate limits precluded multi-sampling at scale, so it is decoded with a *single* sample per query—yet it still outperforms the 14B (Section 6), indicating that model scale dominates decode-time self-consistency at this budget. Prompts are 8-shot for every relation except AWARDWONBY (2-shot), with exemplars from the training split (Brown et al., 2020).

Hardware and implementation. Qwen2.5-14B runs on a single NVIDIA T4 (16 GB) via HuggingFace Transformers in 4-bit (bitsandbytes nf4, float16 compute) (Detmers et al., 2023); quantisation only fits the hardware and does not change the counted parameter budget. Gemma-4-31B is queried over its API from the same machine. Each model processes the full 478-instance validation set.

Baselines. We report two references on the same validation split: the *empty* baseline, which predicts $\emptyset$ for every instance (Section 3), and the *official* organiser baseline, a few-shot prompting system whose per-relation scores we take from the task repository.

Reproducibility. Generation writes a cache of every raw sample; all post-processing—parsing, consistency voting, median aggregation, and the threshold sweep—is a deterministic, GPU-free function of that cache. Every ablation in Section 7 therefore comes from a single generation pass and reproduces without a GPU. We release all code and prompts.

6 Results

Our primary system, **Gemma-4-31B**, scores **0.459** relation-averaged macro- F_1 (**0.490** micro) on validation—versus 0.175/0.203 (empty), 0.273/0.284 (official baseline), and 0.416/0.447 for the smaller Qwen2.5-14B. Figure 3 and Table 3 give the per-relation picture. Gemma-4-31B *beats the official baseline on all six relations* and the 14B on four of six; the gains are largest on COMPANYTRADESATSTOCKEXCHANGE (0.62 $\rightarrow$ 0.72) and HASCAPACITY (0.09 $\rightarrow$ 0.17, nearly doubled), while COUNTRYLANDBORDERSCOUNTRY reaches 0.953. It ties

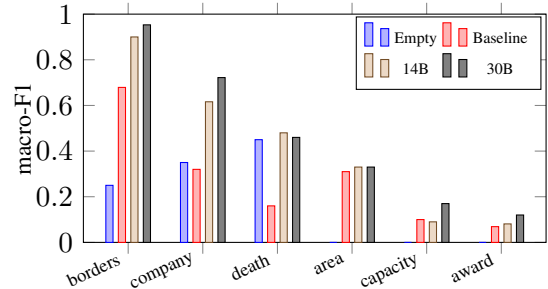


Figure 3: Per-relation validation macro- F_1 : predict-nothing baseline, official baseline, Qwen2.5-14B, and Gemma-4-31B (30B). The 30B beats the official baseline on all six relations.

Relation	Empty	Base	14B	30B
countryLandBorders	0.250	0.679	0.900	0.953
companyTrades	0.350	0.320	0.616	0.722
personCityOfDeath	0.450	0.160	0.480	0.460
hasArea	0.000	0.310	0.330	0.330
hasCapacity	0.000	0.100	0.090	0.170
awardWonBy	0.000	0.069	0.081	0.120
Avg. of relations	0.175	0.273	0.416	0.459
Micro (all inst.)	0.203	0.284	0.447	0.490

Table 3: Validation macro- F_1 by model. “Empty” is the measured predict-nothing baseline, “Base” the official baseline; **30B** is our primary Gemma-4-31B system, 14B is Qwen2.5-14B. Best per row in bold. Gemma-4-31B beats the official baseline on *all six* relations.

the 14B on HASAREA and trails only on PERSON-HASCITYOFDEATH (0.46 vs. 0.48)—the one relation where the single-sample 30B forgoes the abstention voting the 14B uses. Strikingly, the 30B attains its lead *without* self-consistency or abstention, indicating that model scale outweighs those decode-time mechanisms at this budget.

7 Analysis

All ablations are decoded from the same cached samples.

Abstention rescues weak models, refines strong ones. Table 4 isolates abstention on the three null-heavy relations. On a poorly-calibrated 0.5B model the effect is dramatic (0.173 $\rightarrow$ 0.372): without voting it over-answers and falls *below* the empty baseline (0.350). The 14B is already above empty without voting (0.652), so voting adds less (0.652 $\rightarrow$ 0.666) but remains positive. Abstention buys precision: on PERSONHASCITYOFDEATH it moves $P=0.77 \rightarrow 0.81 \rightarrow 0.89$ as θ rises, trading a little recall—it suppresses wrong guesses, not correct answers.

	Off	$\theta=0.5$	$\theta=0.67$
0.5B (avg of 3 null-heavy)	0.173	0.363	0.372
14B (avg of 3 null-heavy)	0.652	0.665	0.666

Table 4: Self-consistency abstention, relation-averaged over the three null-heavy relations only. “Off” decodes a single sample.

Threshold tracks the empty rate; numeric voting is neutral. If abstention is what drives the string relations, the natural question is how its threshold should be set. Figure 4 (left) sweeps θ : the most null-heavy relation, PERSONHASCITY-OFDEATH (45% empty), peaks at a *lower* θ , while the others tolerate aggressive voting—arguing for per-relation thresholds. Figure 4 (right) varies the number of numeric samples: median aggregation is non-monotonic on HASAREA and actually slightly *hurts* HASCAPACITY (single sample 0.10 vs. median-5 0.09). Decode variance is not the bottleneck—the gap is knowledge-bound; the model’s capacity guesses are the right order of magnitude but outside the 5% band (e.g. Spartan Stadium: predicted 37,000, gold 75,005).

AWARDWONBY is recall-capped. The one relation we never beat the baseline on comfortably exposes a different limit. Gold sets average ~ 148 recipients (max 641) but the 14B enumerates only ~ 11 names—its outputs are *not* truncated (it closes the JSON early)—capping recall near $11/148 \approx 0.07$. Listed names are often plausible-but-wrong ($P=0.36$). Parametric recall and willingness to enumerate bound this relation, where a larger in-budget model or multi-step decomposition would help most.

Qualitative behaviour. Table 5 shows representative Gemma-4-31B outputs. The model is exact on well-known facts (PERSONHASCITYOFDEATH, HASCAPACITY) and near-complete on borders, but the two error modes are visible: a numeric near-miss on HASAREA, and on AWARDWONBY a list of globally famous names rather than the actual (obscurer) Stockholm honorary doctors—plausible but wrong, the recall/hallucination limit quantified above.

8 Conclusion

Every component of our system follows from a single metric analysis: abstention is valuable, nu-

meric recall is scarce, and scope grounding curbs over- and under-prediction. With Gemma-4-31B it reaches 0.459 relation-averaged macro- F_1 (0.490 micro), beating the official baseline on all six relations (0.416 with a local 4-bit Qwen2.5-14B), and is fully reproducible from a cached generation pass. That the 30B wins with single-sample decoding, while abstention and self-consistency chiefly help the smaller models, suggests that model scale and decode-time calibration are complementary levers whose balance shifts with model strength.

Limitations

The voting threshold θ is swept on the same validation split it is evaluated on, so per-relation optima are optimistic and not held-out; we report no test numbers yet. Several margins are within plausible decoding noise: AWARDWONBY has only 10 subjects, and we did not measure run-to-run variance of the stochastic decoding ($N=3$, $n=5$), so sub-0.03 deltas (e.g. the PERSONHASCITYOFDEATH +0.03) are indicative only. Being closed-book, the system inherits the base model’s coverage and recency limits and cannot recover facts it never stored, which caps AWARDWONBY recall and rare-venue numeric accuracy. Because API rate limits precluded multi-sample decoding for Gemma-4-31B, its scores are single-sample; pairing the 30B with the self-consistency and abstention voting that help the smaller models is left to future work and may lift the null-heavy and numeric relations further. Finally, the abstention advantage is a property of this task’s macro metric and the empty-as-valid-answer setting, and may not transfer to single-answer or micro-only settings.

Ethics and Generative-AI Declaration

Closed-book KB construction surfaces facts from model parameters that may be stale, biased, or hallucinated; predictions about people (e.g. place of death) can be wrong and should not be treated as authoritative without verification. The abstention mechanism partly mitigates this by emitting an empty answer rather than a confident fabrication when the model is uncertain. The only generative-AI component is the open-weight LM used for the predictions themselves, as the task requires; no proprietary assistant was used to produce results.

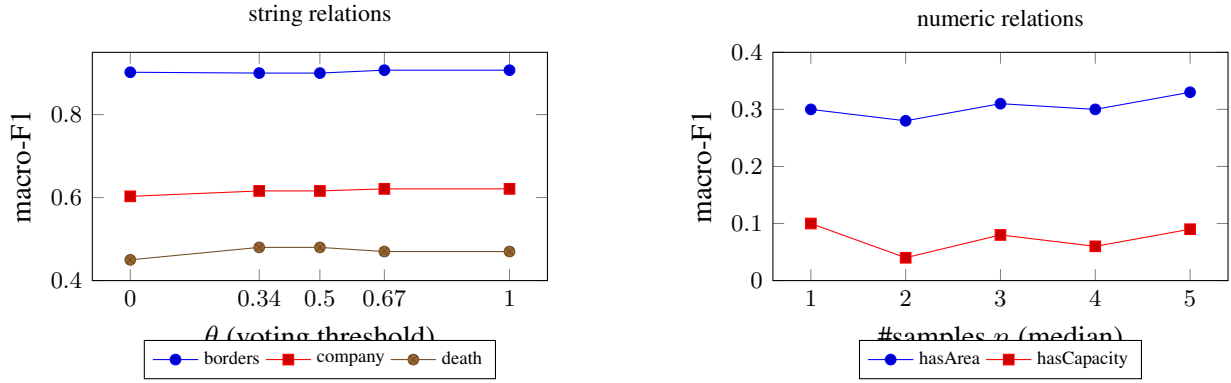


Figure 4: Left: per-relation F_1 vs. voting threshold θ (14B); the most null-heavy relation (PERSONHASCITYOFDEATH, 45% empty) peaks at a lower θ than the others. Right: numeric F_1 vs. number of median samples n ; aggregation is roughly neutral, indicating a knowledge-bound gap.

Relation	Subject	Gold (abbrev.)	Gemma-4-31B output	
borders	Ethiopia	Djibouti, Eritrea, Kenya, Somalia, S. Sudan, Sudan	Djibouti, Eritrea, Kenya, Somalia, S. Sudan	✓ 5/6
death	Vito Acconci	New York City	New York City	✓
company	Bharti Airtel	Bombay Stock Exchange	NSE of India; Bombay Stock Exchange	✓
capacity	Gileno de Carli	5000	5000	✓
area	Lošinj	74.36	123.3	×
award	Stockholm hon. dr	Torvalds, B. Andersson, Gates, ... (40)	Mandela, Tutu, Annan, Yunus, ...	×

Table 5: Qualitative examples: Gemma-4-31B vs. gold. Exact hits on PERSONCITYOFDEATH/HASCAPACITY, near-complete borders; the numeric near-miss (HASAREA) and plausible-but-wrong enumeration (AWARDWONBY) are the knowledge-bound errors of Section 7.

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